Seclusion EDA Findings – CRDC 2017-18

Prepared for: Adhya Putta · October Milestone · Scope: 2017-18 school-level data only

Bottom line

Seclusion is rare (6.6% of schools report any), heavily skewed, and strongly shaped by state reporting context. Model a **binary target (any seclusion)** on schools with at least 30 students, use an **"elevated" tier as a sensitivity target**, and run feature sets **with and without state**. The 2021-22 comparison and race/sex equity checks could not be done because those data were not in the files supplied.

Data completeness

- 97,632 schools; 91,013 (93.2%) are analysis-ready. All keys, flags and derived fields are internally consistent, and the cleaning log reconciles with the NaNs in the schools file.
- Seclusion is missing for 6,619 schools (6.8%) because any reserved-code component blanks the total. 226 of them had an observed component above zero. Dropped schools are more often special-education schools (5.6% vs 2.0%), so the analysis file under-represents high-risk schools.
- `sch_virt_ind` and `enr_nonbinary` are 100% empty.

What the data show

- **Distribution:** 93.4% of schools report zero. Among the 5,975 that report any, the median is 5 instances, the 90th percentile 54, the maximum 1,252. The top 100 schools hold 30% of all instances.
- **State dominates:** Wisconsin, Iowa, Maine, Connecticut and Kansas have 23-33% of schools reporting any; California, Texas, Nevada, New York and Pennsylvania are near or under 1%; Puerto Rico reports none. State alone gives ROC-AUC 0.81. Adding state to a baseline model lifts average precision from 0.15 to 0.34. This reflects state law and reporting practice as well as real differences.
- **Concentration:** only 12.9% of districts have any school reporting seclusion; 89 districts hold half of all instances.

- **School type:** special-education schools 16.6% any (vs 6.4%) and 23.5% of instances; justice facilities 7.9%; alternative 2.0%, charter 2.8%, magnet 4.7%. Elementary 8.9%, middle 5.1%, high 2.9%. 93% of positive schools are ordinary schools.
- **Population:** %IDEA is the clearest positive correlate (2.1% any at under 5% IDEA, 8.5% at 15-20%). %504 is negative at high levels; %EL is inverted-U. All rank correlations with the outcome are weak (|rho| ⩽ 0.07).
- **Staffing:** non-monotonic and confounded. After adjustment, psychologists are slightly positive and security guards negative; counselors, nurses and social workers add little. Security and support staffing may respond to behaviour problems, so treat these as associations.
- **Small schools distort rates:** schools under 30 students are 3.5% of schools but 31% of those with more than 100 instances per 100 students.
- **Equity (disability only):** students with IDEA are 13.2% of students but 85.9% of instances: 1.94 instances per 100 students vs 0.046 for students without a disability (about 42x). These are events, not unique students.

Recommended target definition

- **Primary, `any_seclusion`:** seclusion_instances ⩾ 1. Prevalence 6.7% in the modeling frame.
- **Secondary, `elevated_seclusion`:** seclusion_instances ⩾ 5 and seclusion_rate ⩾ 1 per 100 students. Prevalence 3.1% (2,710 schools). Five instances covers 96% of all instances; the rate floor stops large schools qualifying on volume alone.
- **Modeling frame:** analysis-ready rows with enrollment ⩾ 30 (drops 3,230 schools), excluding the 68 FTE-outlier rows. Use grouped cross-validation by district. For any rate-based work, add shrinkage (about 30-100 pseudo-students) or use counts with a log-enrollment offset.
- **Not recommended:** raw seclusion rate or count as a regression target.

Candidate feature shortlist

- **Include:** log enrollment, grade band, special-education / alternative / charter / justice flags, %IDEA, %504, %EL, psychologists and social workers per 100 students, log district enrollment.
- **Optional:** magnet flag, counselors and nurses per 100, teachers per 100, security staff per 100 (reverse-causality caution).

- **Second model only:** state (or region) as a categorical.
- **Drop as redundant:** raw teacher and counselor FTE (correlated 0.90 / 0.53 with enrollment), `support_fte_per_100`, `lea_schools` (0.97 with district enrollment).
- **Exclude as leakage:** the three seclusion component counts, instances and rate. Other discipline and restraint measures were not in the cleaned files, so leakage could not be tested directly; keep them out of the baseline.

Data issues for the cleaning owners

- **High:** impossible FTE values remain (19,048.6 psychologists at one school; 102,579 guards); the outlier flag covers 68 rows and flagged rows are still in the analysis file.
- **High:** the seclusion total is blanked when any component is reserved, discarding 226 known lower bounds.
- **High:** extreme counts in ordinary schools need verification against CRDC (e.g. Claiborne Pell Elementary, RI: 829 instances / 898 students; 155 analysis rows have more instances than students).
- **High:** race and sex of secluded students, and other discipline and restraint columns, were not carried through cleaning.
- **High:** `sch_virt_ind` is entirely empty.
- **Medium:** `seclusion_rate` is really instances per 100 students and can exceed 100; the small-enrollment flag (under 10) is too low for rates and is not applied; Puerto Rico and several states report almost no seclusion.
- **Low:** six flags never fire, `enr_nonbinary` is empty, one quality-log row has no school key, and `fte_leo` / `fte_guards` have 1.7% / 2.3% missing.

The full list with row counts is in `data_issues_201718.csv`.

Open items

1. Supply 2021-22 (and 2020-21) cleaned files to complete the cross-year comparison.
2. Carry race, sex and the other discipline and restraint columns through cleaning, then rerun the equity and leakage sections.
3. Team decision: confirm the minimum enrollment (30) and whether the headline model includes state.